



Rajiv Gandhi Institute of Petroleum Technology Jais, Amethi

An Institution of National Importance, Government of India

Design & Innovation Centre (DIC) — Problem Statements & Faculty Mentors

Prob. No.	Problem Statement	Faculty Mentor
1	Development of Eco-Efficient Dip Coating Technologies for Corrosion Protection and Zero-Liquid-Discharge Waste Handling	Dr. Deepak Dwivedi
2	Nanostructured Coatings for Enhanced Corrosion Resistance in Harsh Environments	Dr. Deepak Dwivedi
3	Design of Safe, Scalable, and Economically Viable E-Waste Recycling Systems with Resource Recovery Integration	Dr. Karan Malik
4	Urban Mining: Extracting Precious Metals from Electronic Waste	Gunjan Kumar Agrahari
5	Development of High-Return Biogas Slurry Processing Systems for CO ₂ Adsorption, Energy, Fertilizer, and Water Reuse	Tushar Sharma
9	Next-Generation MSW Segregation Systems: Low-Noise, Low-Vibration, and High-Throughput Design	Hreetabh Kishore
13	Sustainable Plastic Waste Recycling Technologies for High-Value Product Generation using Product Manufacturing Techniques	Hreetabh Kishore
13	Sustainable Plastic Waste Recycling Technologies for High-Value Product Generation using Product Manufacturing Techniques	Naveen Mani Tripathi
17	Development of Voice-to-Text Software for Meeting Documentation	Dosa Office
19	Development of Techniques / Systems for CO ₂ Removal from Biogas	Koushik Guha Biswas
20	Development of an Activity Planning and Documentation Software for Dosa Office	Sajal Agarwal
20	Development of an Activity Planning and Documentation Software for Dosa Office	Koushik Guha Biswas
24	CO ₂ Sequestration in Geological Formations	Dr. Shivanjali Sharma
27	CCUS Technologies	Tushar Sharma
31	Wellbore Instabilities and its Mitigation	Dr. Piyush Sarkar
32	AI/ML Technology for Seismic Data Processing and Interpretation	Prof. Satish Kumar Sinha
38	Reservoir Geomechanics and Porous Media Characterization	Dr. Piyush Sarkar
40	Flow Assurance in Tubing and Pipelines	Dr. Shivanjali Sharma
40	Flow Assurance in Tubing and Pipelines	Dr. Naveen Mani Tripathi
41	Mitigation of Asphaltene and Wax Deposition	Dr. Shailesh Kumar
44	Water Shut Off	Prof. A K Choubey
45	Drilling / Completion / Workover / Fracturing Fluid Design	Dr. Shailesh Kumar
52	Digitalization / AI / ML for Oil and Gas Industry	Prof. Satish Kumar Sinha



Prob. No.	Problem Statement	Faculty Mentor
57	Development of Intelligent Nanoparticle-Enhanced Fluids for Optimized Oil Recovery and Improving CO ₂ Sequestration Potential	Dr. Shivanjali Sharma
58	Advanced Cement Formulations for High-Temperature and High-Pressure Challenging Downhole Conditions	Dr. Amit Saxena
59	Development of Responsive Drilling Fluids for Improved Drilling and Wellbore Stability	Dr. Amit Saxena
62	Hydrate Formation Modeling in Gas Pipelines	Prof. Amit Ranjan
69	Low-Cost Catalytic Pyrolysis of Plastic Waste into Fuels	Dr. Arshad Aijaz
71	Hexene-1 Production through Controlled Trimerization Approach	Dr. Omvir Singh
86	Green Hydrogen Generation	Milan Kumar
86	Green Hydrogen Generation	Dr. Supriya
87	Redox Flow Battery	Milan Kumar
88	Sodium-Ion Battery	Dr. Supriya
90	CO ₂ to Chemicals	Dr. Omvir Singh
103	Mechanical Issues of Wellbore Instabilities	Prof. Amit Ranjan
104	AI/ML Techniques for Casting, Welding, and Industrial Manufacturing	Hreetabh Kishore
108	Design of Metal Oxide Semiconductor (MOS) using Indigenous Nanomaterial	Dr. Riya Wadhwa
128	Cybersecurity Solutions for Refinery OT/IT Systems	Dr. Saptarshi Ghosh
128	Cybersecurity Solutions for Refinery OT/IT Systems	Dr. Gobinda Rakshit
130	AI/ML Image Analysis for Corrosion & Crack Detection	Dr. Gargi
130	AI/ML Image Analysis for Corrosion & Crack Detection	Dr. Ankur
135	UAV, CCTV, IoT, and Balloon-Based Surveillance Systems	Dr. Akash Yadav
138	LLM Digital Process Engineer – Conversational Assistant	Dr. Santosh Mishra
141	Regulatory Compliance Checker using NLP	Dr. Santosh Mishra
143	Energy Forecasting and Demand Prediction (ARIMA, ML/DL, Bayesian)	Dr. Shrawan Kumar Trivedi
144	Energy Trading and Market Analytics	Rohit Bansal
144	Energy Trading and Market Analytics	Manoj K Rajpoot
146	Data Assimilation for Real-Time Reservoir Management	Dr. Sudeep Kundu



Prob. No.	Problem Statement	Faculty Mentor
147	Algorithmic & High-Frequency Trading (HFT) using AI/ML	Dr. Manoj K Rajpoot
154	Prediction of Energy Storage Capability by Machine Learning	Dr. Soniya Dhama
155	Data Driven Classification Approach for Specific Capacitance Prediction of Energy Storage Materials	Dr. Amarish Dubey
156	Flexible Display for Better System Control in Energy Sector	Dr. Amarish Dubey
162	Multilevel Converters for Renewable Integration in Oil & Gas Operations	Dr. Vijay Kumar Singh
163	Modular Power Converters for Scalable Backup and Hybrid Energy Systems	Dr. Vijay Kumar Singh
163	Modular Power Converters for Scalable Backup and Hybrid Energy Systems	Dr. Saurabh Pandey
175	CO ₂ to Value Added Chemicals through Electrolysis	Dr. Karan Malik
177	Electrocatalytic Upcycling of Plastic Waste into High-Value Chemicals	Dr. Malaya Kumar Sahoo
177	Electrocatalytic Upcycling of Plastic Waste into High-Value Chemicals	Dr. Arshad Aijaz
178	Energy Conversion and Storage: Materials Development and Device Optimization	Dr. Malaya Kumar Sahoo
179	Photocatalytic Materials for Energy and Environmental Applications	Shikha Singh
179	Photocatalytic Materials for Energy and Environmental Applications	Vipin Amoli
180	Artificial Photosynthesis & Solar Fuel Generation	Shikha Singh
181	Molecular & Hybrid Catalysts, Water Splitting, CO ₂ Photoreduction, Integrated Solar Reactors	Debashis Panda
182	Biomass Conversion Technologies for CCUS and Energy (Hydrogen, Fuel) Production	Debashis Panda
183	Lignocellulosic Biomass Valorization	Dr. Vasu Chaudhary
185	Thermo-/Photo-/Bio-Catalysis	Arvind Singh
187	Next-Gen Energy Materials for Sensors, Storage, & Conversion	Vipin Amoli
188	2D & Quantum Materials	Dr. Tanmoy Ghosh
191	Thermoelectric Materials	Dr. Tanmoy Ghosh
194	AI for Predictive Maintenance in Energy Systems	Dr. Shrawan Kumar Trivedi
197	Occupational Health Monitoring, Chemical and Environmental Exposure Risks	Dr. Akash Yadav
201	AI/ML Models for Reservoir Simulation	Dr. Sudeep Kundu
205	AI/ML Model for Reservoir Characterization	Pallabi Saikia



Prob. No.	Problem Statement	Faculty Mentor
208	Multimodal AI for Asset Health Monitoring	Pallabi Saikia
210	Energy Economics & Policy Modeling: Scenario Modeling for Net Zero India	Rohit Bansal

Total Entries: 75 | Unique Problems: 65 | Students must submit preferences using the official Google Form. | All allocations are subject to faculty approval.